# 梯度运算的常用推论表

设多元函数 $u = u(x, y, z)$ ， $v = v(x, y, z)$ 均有一阶连续偏导数，梯度定义为 $\mathrm{grad}, F = \left( \frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right)$ ，以下按序号逐一证明：

### (1) 证明 $\mathrm{grad}(\boldsymbol{au+bv}) = \boldsymbol{a}\mathrm{grad}\boldsymbol{, u} + \boldsymbol{b}\mathrm{grad}\boldsymbol{, v}$ （$a, b$ 为常数）

根据梯度定义展开左边：

$$\mathrm{grad}(au+bv) = \left( \frac{\partial (au+bv)}{\partial x}, \frac{\partial (au+bv)}{\partial y}, \frac{\partial (au+bv)}{\partial z} \right)$$

由偏导数的线性运算法则：

$$\frac{\partial (au+bv)}{\partial x} = a\frac{\partial u}{\partial x} + b\frac{\partial v}{\partial x}$$
$$\frac{\partial (au+bv)}{\partial y} = a\frac{\partial u}{\partial y} + b\frac{\partial v}{\partial y}$$
$$\frac{\partial (au+bv)}{\partial z} = a\frac{\partial u}{\partial z} + b\frac{\partial v}{\partial z}$$

因此：

$$\mathrm{grad}(au+bv) = \left( a\frac{\partial u}{\partial x} + b\frac{\partial v}{\partial x}, a\frac{\partial u}{\partial y} + b\frac{\partial v}{\partial y}, a\frac{\partial u}{\partial z} + b\frac{\partial v}{\partial z} \right) = a\,\mathrm{grad}\,u + b\,\mathrm{grad}\,v$$

得证。

### (2) 证明 $\mathrm{grad}(\boldsymbol{uv}) = \boldsymbol{v}\mathrm{grad}\boldsymbol{, u} + \boldsymbol{u}\mathrm{grad}\boldsymbol{, v}$

展开左边梯度：

$$\mathrm{grad}(uv) = \left( \frac{\partial (uv)}{\partial x}, \frac{\partial (uv)}{\partial y}, \frac{\partial (uv)}{\partial z} \right)$$

由乘积的偏导数法则（莱布尼茨法则）：

$$\frac{\partial (uv)}{\partial x} = v\frac{\partial u}{\partial x} + u\frac{\partial v}{\partial x}$$
$$\frac{\partial (uv)}{\partial y} = v\frac{\partial u}{\partial y} + u\frac{\partial v}{\partial y}$$
$$\frac{\partial (uv)}{\partial z} = v\frac{\partial u}{\partial z} + u\frac{\partial v}{\partial z}$$

因此：

$$\operatorname{grad}(uv)=\left(v\frac{\partial u}{\partial x}+u\frac{\partial v}{\partial x},v\frac{\partial u}{\partial y}+u\frac{\partial v}{\partial y},v\frac{\partial u}{\partial z}+u\frac{\partial v}{\partial z}\right)=v\operatorname{grad}u+u\operatorname{grad}v$$

得证。

### (3) 证明 $\operatorname{grad}\boldsymbol{\frac{u}{v}}=\boldsymbol{\frac{v\mathrm{grad}, u-u\mathrm{grad}, v}{v^2}}$ （$v\neq 0$）

展开左边梯度：

$$\operatorname{grad}\frac{u}{v}=\left(\frac{\partial\left(\frac{u}{v}\right)}{\partial x},\frac{\partial\left(\frac{u}{v}\right)}{\partial y},\frac{\partial\left(\frac{u}{v}\right)}{\partial z}\right)$$

由商的偏导数法则：

$$\frac{\partial\left(\frac{u}{v}\right)}{\partial x}=\frac{v\frac{\partial u}{\partial x}-u\frac{\partial v}{\partial x}}{v^2}$$
$$\frac{\partial\left(\frac{u}{v}\right)}{\partial y}=\frac{v\frac{\partial u}{\partial y}-u\frac{\partial v}{\partial y}}{v^2}$$
$$\frac{\partial\left(\frac{u}{v}\right)}{\partial z}=\frac{v\frac{\partial u}{\partial z}-u\frac{\partial v}{\partial z}}{v^2}$$

因此：

$$\operatorname{grad}\frac{u}{v}=\frac{1}{v^2}\left(v\frac{\partial u}{\partial x}-u\frac{\partial v}{\partial x},v\frac{\partial u}{\partial y}-u\frac{\partial v}{\partial y},v\frac{\partial u}{\partial z}-u\frac{\partial v}{\partial z}\right)=\frac{v\operatorname{grad}u-u\operatorname{grad}v}{v^2}$$

得证。

### (4) 证明 $\mathrm{grad}\boldsymbol{, f(u)=f'(u)}\mathrm{grad}\boldsymbol{, u}$ （$f$ 可导）

设 $F=f(u)$，由复合函数偏导数的链式法则：

$$\frac{\partial F}{\partial x}=f'(u)\frac{\partial u}{\partial x}$$
$$\frac{\partial F}{\partial y}=f'(u)\frac{\partial u}{\partial y}$$
$$\frac{\partial F}{\partial z}=f'(u)\frac{\partial u}{\partial z}$$

因此：

$$\operatorname{grad}f(u)=\left(f'(u)\frac{\partial u}{\partial x},f'(u)\frac{\partial u}{\partial y},f'(u)\frac{\partial u}{\partial z}\right)=f'(u)\operatorname{grad}u$$

得证。

（注：文档部分内容可能由 AI 生成）